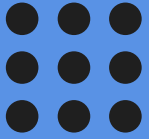




มหาวิทยาลัยมหิดล
Mahidol University



MTMI 202 Basic Microbiology: 2567/1

October 15, 2024

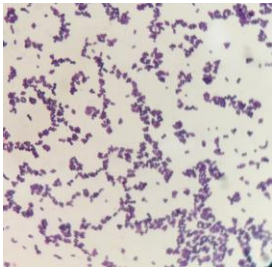
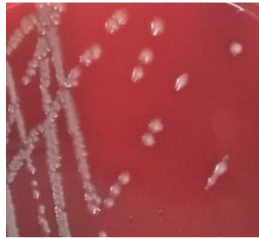
Biochemical Test for Bacterial Identification

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Clinical Microbiology and Applied Technology,
Medical Technology, Mahidol University



Bacterial Identification



Colony morphology

Gram & Shape & Arrangement

- Gram positive cocci in cluster or in chain
- Gram negative bacilli

Biochemical Characteristic

Learning outcome

Able to explain principle process and report biochemical tests

Classification of biochemical tests

1.	Metabolism of carbohydrate (CHO) ➡ Acid
2.	Carbon-source substrate
3.	Metabolism of amino acid ➡ Alkaline
4.	Enzyme production
5.	Tolerance tests
6.	Susceptibility tests
7.	Miscellaneous tests

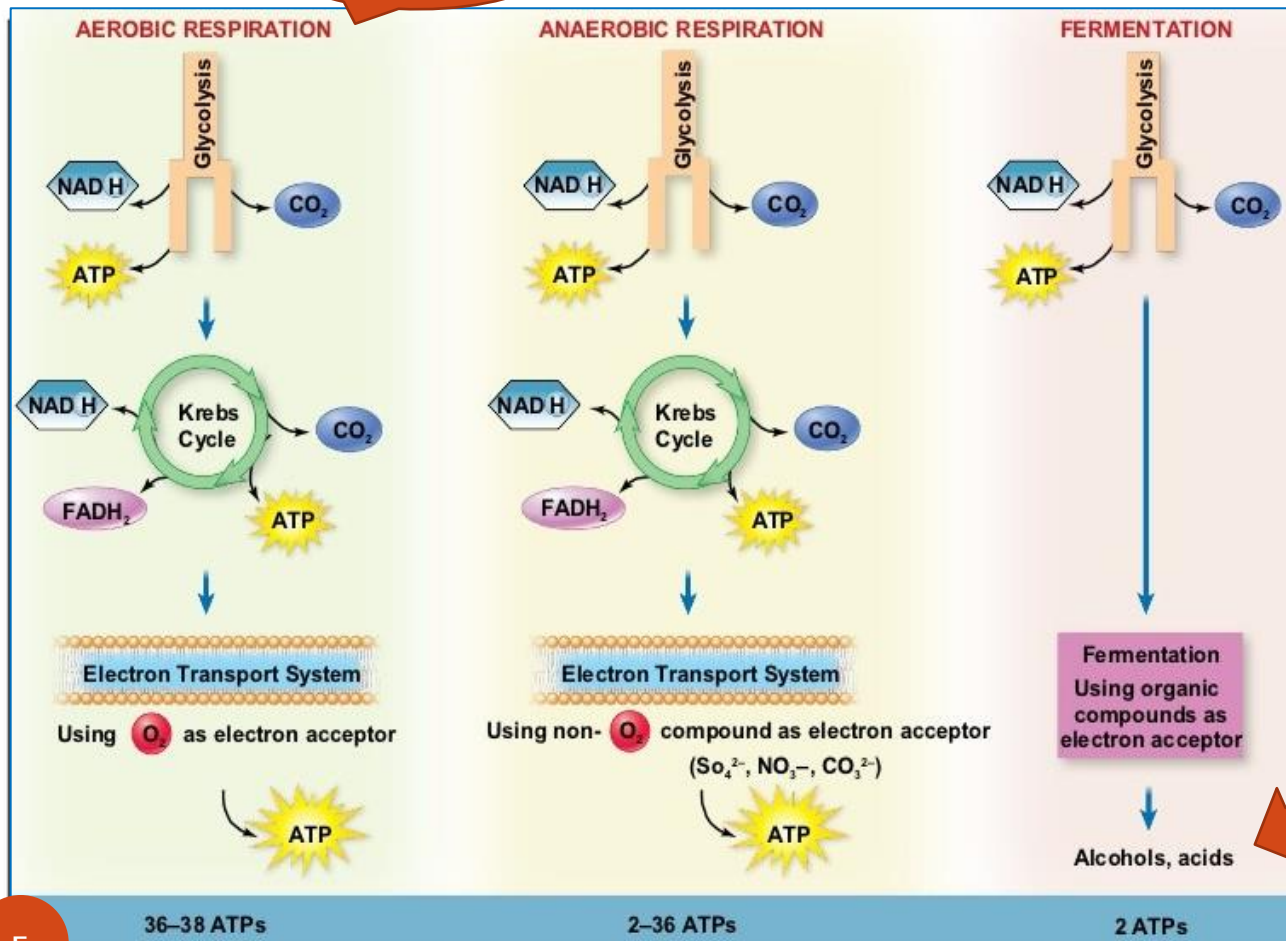
Indicator

	Acidity	←	→	Alkalinity
• Bromothymol blue			 (pH 6.0-7.6)	
• Bromocresol purple			 (pH 5.2-6.8)	
• Andrade's			 (pH 5.5-7.2)	
• Phenol red			 (pH 6.6-8.0)	
• Methyl red			 (pH 4.8-6.0)	

Uninoculated media

1. Metabolism of carbohydrate

Oxidation



Bacteria can be classified based on their ability to ferment carbohydrates:

- Fermentative bacteria
- Non-fermentative bacteria (NFB) or Oxidative bacteria

1. Metabolism of carbohydrate

1.1 Oxidation-Fermentation (OF) test: Glucose O/F test

- For gram positive cocci
- For gram negative bacilli, especially NFB

1.2 Triple Sugar Iron agar (TSI)

1.3 Sugar fermentation test

1.4 Specific end products from CHO fermentation

- Methyl Red (MR)
- Voges-Proskauer (VP)

1.5 Sugar oxidation test

- For NFB
- For *Neisseria* spp. → Cystine Tryptic Agar-Sugar (CTA-Sugar)

1.1 Oxidation-Fermentation (OF) test

Glucose O/F test:

Glucose Bacilli Carbo Y₂ if Y₂ is not

❖ **Purpose:** To determine how bacteria utilize CHO

❖ **Principle:** 1. O (oxidation, aerobic cond.)

Glucose → Pyruvate → → weak acid

2. F (fermentation, anaerobic cond.)

Glucose → Pyruvate → mixed acid, alcohol, H₂, CO₂, acetoin

- **Main components:** basal medium, **Glucose**, agar -> semi-solid *
- **Indicator:** Bromocresol purple → Gram positive cocci (*Staph.* & *Micrococcus*)

Bromothymol blue → Gram negative bacilli, especially NFB

➤ **Interpretation:**

Positive: Color change to **Yellow**

Negative: Color not change



1.1 Oxidation-Fermentation (OF) test

Glucose O/F test: Gram positive cocci (Staph. & Micrococcus)

➤ Bromocresol purple

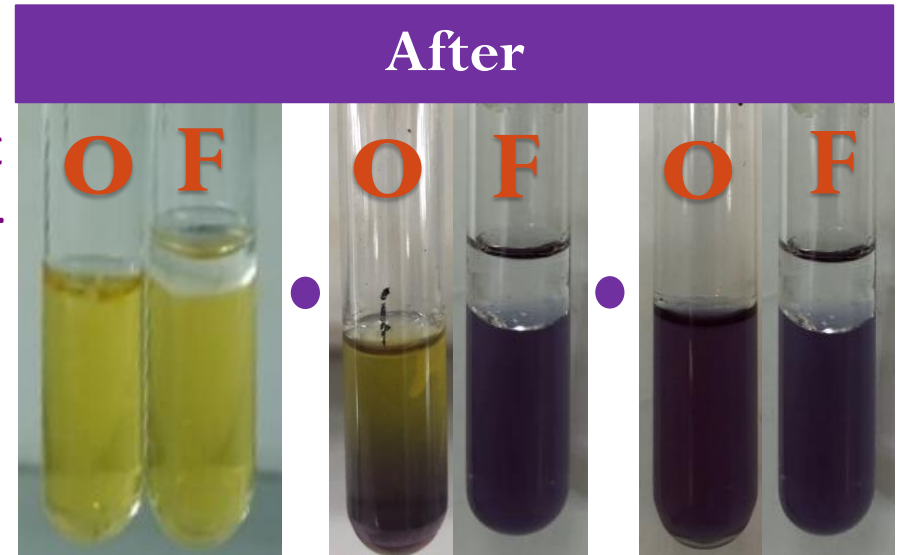


Uninoculated



Inoculated

35-37 °C
24-48 hr.



+/+

- Fermenter
- Facultative
anaerobe

+/-

- Oxidizer

-/-

- Non-
saccharolytic
bacteria



1.1 Oxidation-Fermentation (OF) test

Glucose O/F test: Gram negative bacilli, especially NFB

➤ Bromothymol blue

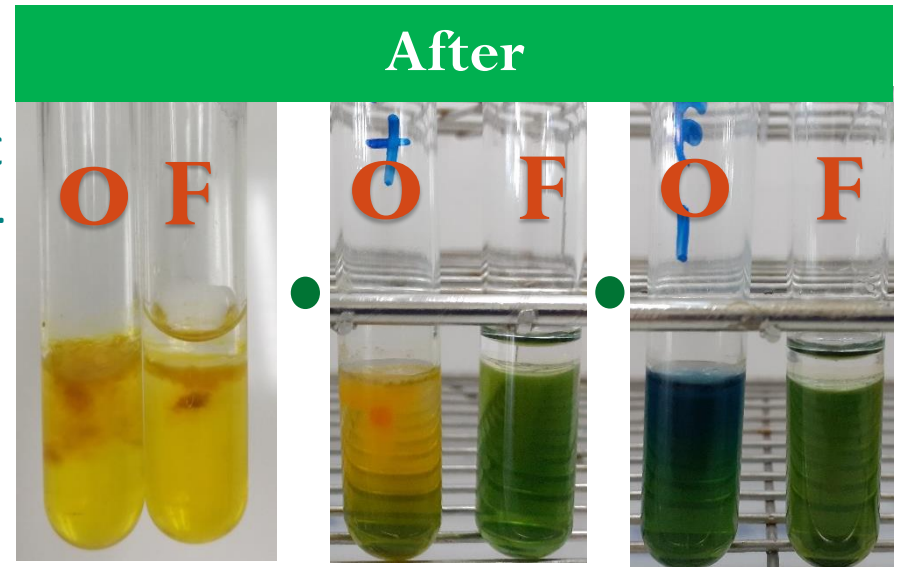


Uninoculated



Inoculated

35-37 °C
24-48 hr.



+/+

- Fermenter
- Facultative
anaerobe

+/-

- Oxidizer
- Non-
fermentative
bacteria (NFB)

-/-

- Non-
saccharolytic
bacteria

1.2 Triple Sugar Iron (TSI) agar test

Triple Sugar Iron (TSI): For **gram negative bacilli**, etc.

Glucose, lactose, sucrose / H_2S

❖ **Purpose:** To determine the ability of bacteria to utilize glucose, lactose, and sucrose, and its ability to produce Hydrogen sulfide (H_2S)

❖ **Principle:** 1. CHO $\rightarrow$ glucose $\rightarrow$ pyruvate $\rightarrow$ mixed acid, alcohol, H_2 , CO_2 , acetoin / weak acid

2. Sodium thiosulfate $\rightarrow H_2S$

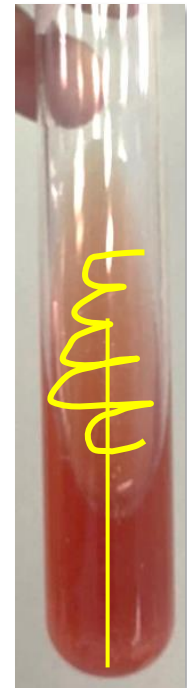
H_2S + ferric ammonium citrate $\rightarrow$ ferrous sulfide
(black precipitation)

➤ **Main components:**

- Basal medium: Peptone $\rightarrow$ N source for bacterial growth
- **Triple Sugar: Glucose (0.1%) + Lactose (1%) + Sucrose (1%)**
- Sodium thiosulfate $\rightarrow$ Source for S reduction
- Ferric ammonium citrate $\rightarrow H_2S$ detection
- Agar $\rightarrow$ solid agar $\rightarrow$ Gas detection

➤ **Indicator: Phenol red**

➤ **Incubation:** at 35-37 °C 18-24 hr.



1.2 Triple Sugar Iron (TSI) agar test

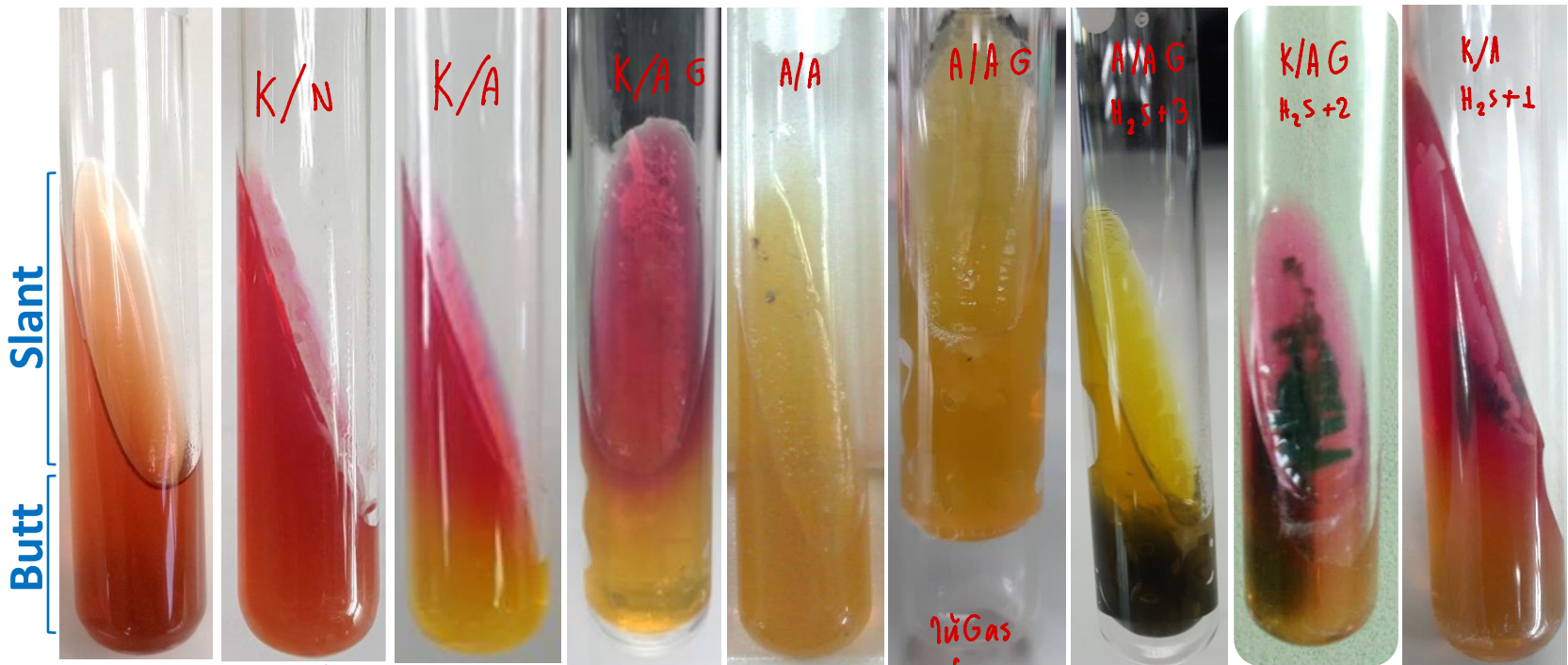
➤ Interpretation: (Indicator: **Phenol red**)

1. Fermentation of Triple Sugar: **Glucose** : **Lactose** : **Sucrose** = 1:10:10

Slant/Butt: **A = Acid**, **N = Neutral**, **K = Alkaline**

2. G = Gas

3. H₂S = Hydrogen sulfide (+1 to +3)



non-fer

fer

fer

fer
ใช้ น้ำตาล 7 ± ตัว

fer
ใช้ น้ำตาล 7 ± ตัว

Report ?

1.3 Sugar fermentation test

Sugar fermentation test: For fermentative bacteria

၅၅၇ Bac Fermentaria ၆၂၂

❖ **Purpose:** To detect sugar fermentation by bacteria

❖ **Principle:** CHO → glucose → pyruvate → mixed acid, alcohol, H₂, CO₂, acetoin

➤ **Sugar:** glucose, lactose, sucrose, maltose, mannitol, trehalose, etc.

➤ **Main components:** Basal medium, Mannitol

➤ **Indicator:** Andrade's

➤ **Interpretation:**

Positive: color change to Pink

Negative: not change



Before



Uninoculated

35-37 °C
18-24 hr.



After



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၆၅၇
၆၂၂

1.3 Sugar fermentation test

Gas from Glucose: For fermentative bacteria

gas & Gas formation ferment glucose

- ❖ **Purpose:** To detect gas production by glucose fermentation
- ❖ **Principle:** Glucose $\rightarrow$ pyruvate $\rightarrow$ mixed acid, alcohol, H_2 , CO_2 , acetoin
 - **Main components:** basal medium, Glucose, Durham tube
 - **Indicator:** Andrade's
 - **Interpretation:**

Positive: color change to Pink
and Gas in Durham tube

Negative: color change to Pink
and no Gas in Durham tube

Before

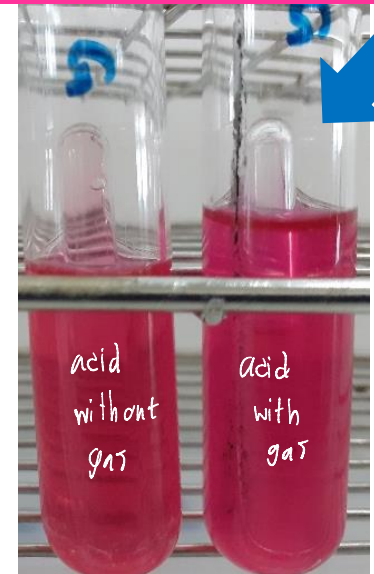


Uninoculated

35-37 °C
18-24 hr.



After



—

+

1.4 Specific end products from sugar fermentation

Methyl Red (MR): For fermentative gram negative bacilli

❖ **Purpose:** To detect stable mixed acid as an end products from glucose fermentation

❖ **Principle:** Glucose → pyruvate → stable mixed acid

➤ **Main components:** Basal medium, **Glucose**

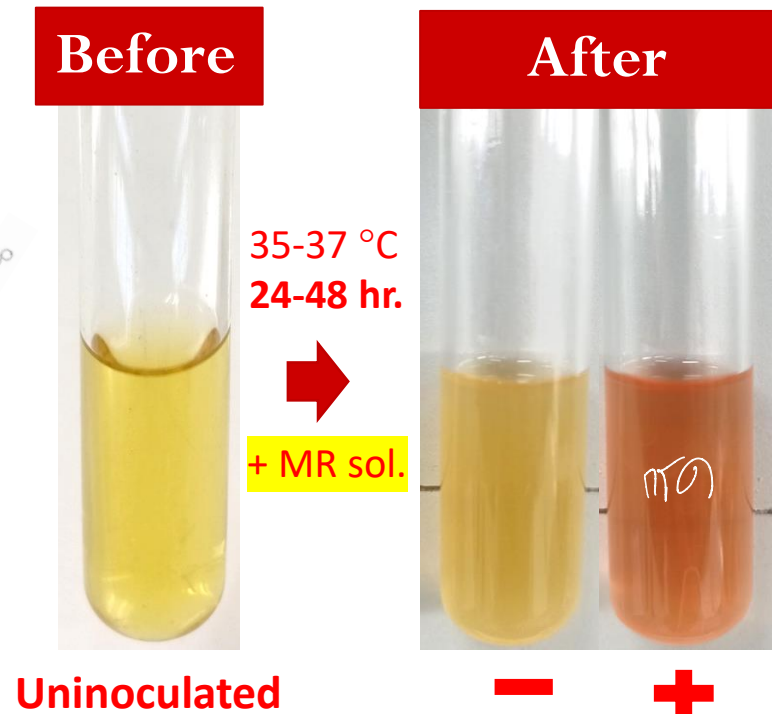
➤ **Indicator:** **Methyl red**

➤ **Reagent test:** **Methyl red (MR) solution**
(2-3 drop)

➤ **Interpretation:**

Positive: color change to **Red**

Negative: not change



1.4 Specific end products from sugar fermentation

Voges-Proskauer (VP): For fermentative gram negative bacilli

❖ **Purpose:** To detect acetoin as an end products from glucose fermentation

❖ **Principle:** Glucose $\rightarrow$ pyruvate $\rightarrow$ acetoin

➤ **Main components:** Basal medium, **Glucose**

➤ **Reagent test:** 5% α -naphthol solution (6 drops)
40% KOH (2 drops)

➤ **Procedure:** Mix vigorously and wait for 10-15 min

➤ **Interpretation:**

Positive: color change to **Red**

Negative: not change



Before

35-37 °C
24-48 hr.

+ 5% α -naphthol
+ 40% KOH

Uninoculated

After



—

+

1.5 Sugar oxidation test

Sugar oxidation test : For **non-fermentative bacilli (NFB)**

- ❖ **Purpose:** To detect sugar oxidation by bacteria
- ❖ **Principle:** CHO → glucose → pyruvate → weak acids
- **Sugar:** glucose, maltose, lactose, xylose
- **Main components:** basal medium, Sugar, agar -> Semi-solid
- **Indicator:** Bromothymol blue
- **Procedure:** Always included Control tube *
- **Interpretation:**
 - Positive:** color change to **Yellow**
 - Negative:** not change or blue (compared with control)

Control



○

No sugar

○

N source only

Test



○

Sugar

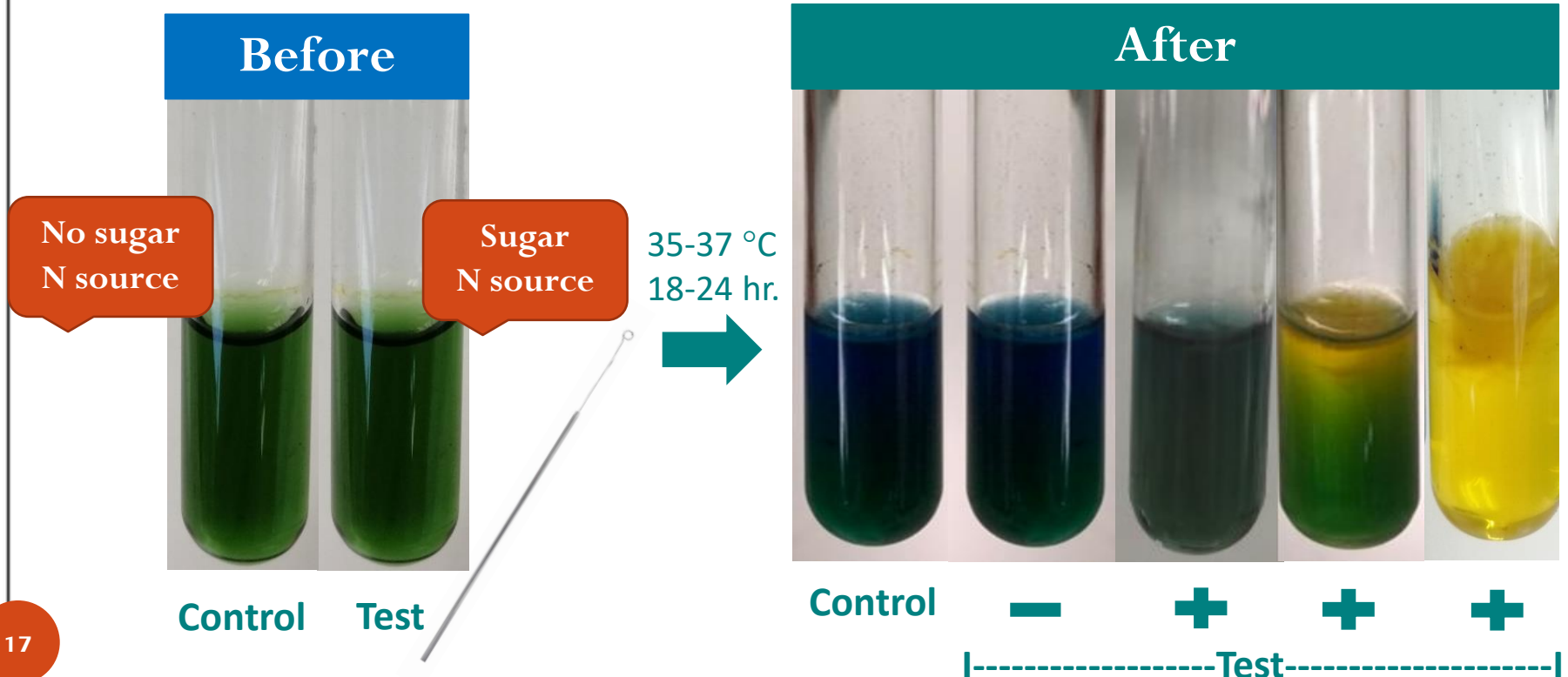
○

N source

1.5 Sugar oxidation test

Sugar oxidation test : For **non fermentative bacilli (NFB)**

- Procedure: Always included **Control tube**
- Interpretation: **Bromothymol blue** Positive: color change to **Yellow**
Negative: not change or blue (compared with control)



1.5 Sugar oxidation test

Cystine Tryptic Agar (CTA)-Sugar test: For *Neisseria* spp.

❖ **Purpose:** To detect sugar oxidation by bacteria

❖ **Principle:** CHO → glucose → pyruvate → weak acids

➤ **Sugar:** glucose, maltose, lactose, sucrose

➤ **Main components:** basal medium, Sugar, agar -> solid agar

➤ **Indicator:** Phenol red

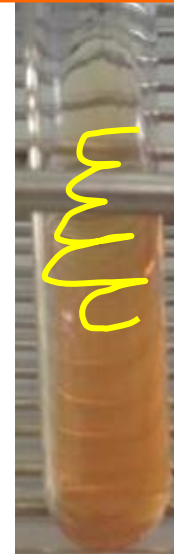
➤ **Interpretation:**

Positive: color change to **Yellow**

Negative: not change or pink-red



Before



Uninoculated

35-37 °C
18-24 hr.



After



—

+

2. Carbon source utilization

2.1 Citrate test

2.2 Malonate test

“Avoid using a large inoculum that may cause false positive result”

2.1 Citrate test

Citrate test: For **gram negative bacilli**, etc.

❖ **Purpose:** To determine the ability of bacteria to utilize citrate as a source of energy

❖ **Principle:** Use sodium citrate (C source)



Cell growth → use $\text{NH}_4\text{H}_2\text{PO}_4$ (N source) → NH_3 (alkaline)

➤ **Main components:** Simmon's Citrate Agar
(sodium citrate, ammonium salts
, agar → solid agar)

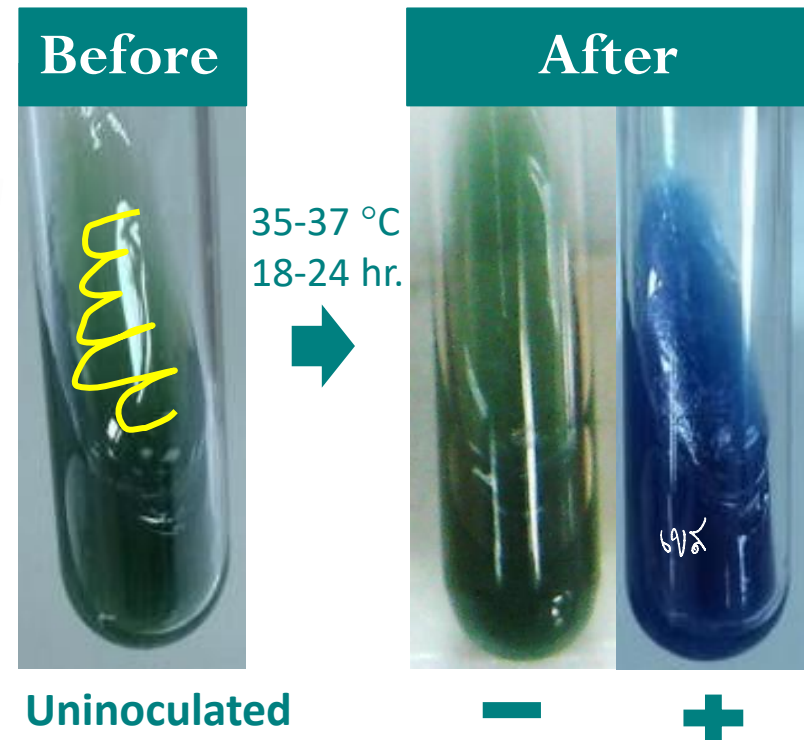
➤ **Indicator:** **Bromothymol blue**

➤ **Interpretation:**

Positive: Growth, color change to **Blue**

Negative: No growth

*Large inoculum → FP



2.2 Malonate test

Malonate test: For gram negative bacilli

❖ **Purpose:** To determine the ability of bacteria to utilize malonate as a source of energy

❖ **Principle:** Use sodium malonate (C source)



Cell growth → Use $(\text{NH}_4)_2\text{SO}_4$ (N source) → NaOH (alkaline)

➤ **Main components:** Malonate-PD broth
*(combination test)

:sodium malonate, ammonium salts

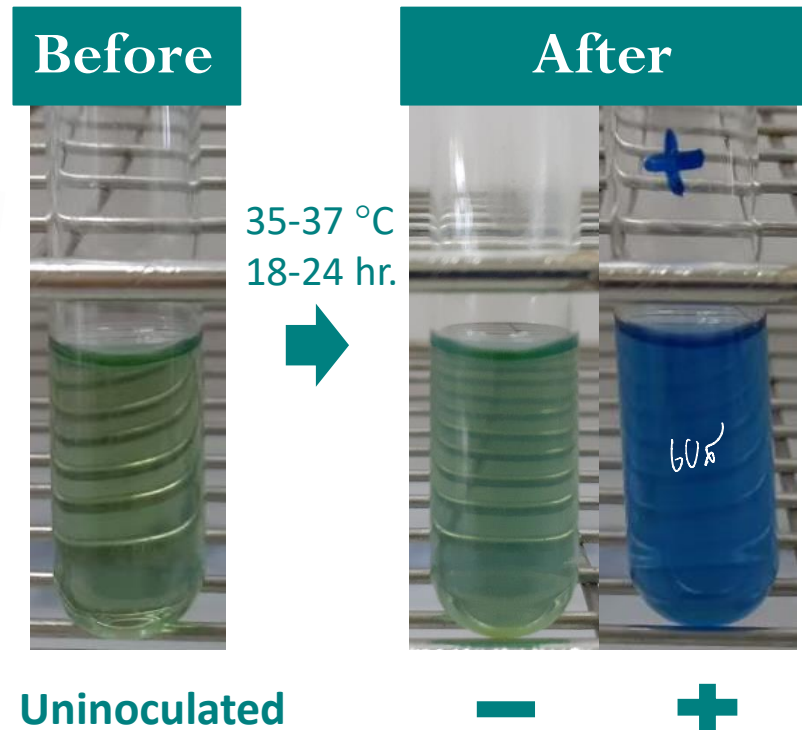
➤ **Indicator:** Bromothymol blue

➤ **Interpretation:**

Positive: color change to Blue

Negative: not change

*Large inoculum → FP



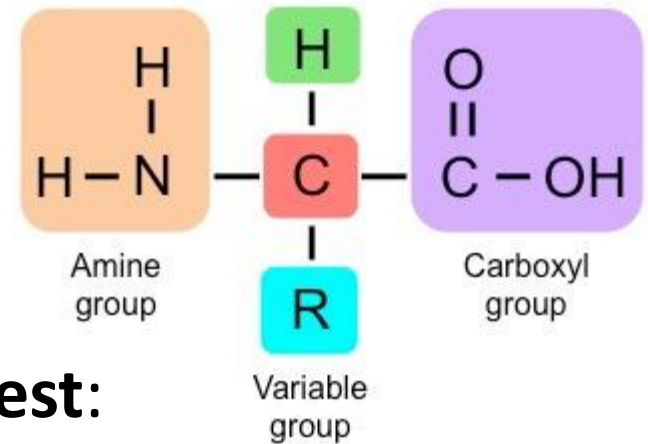
3. Metabolism of amino acid

3.1 Deaminase test:

- Phenylalanine deaminase (PD)
- Indole test

3.2 Decarboxylase/Dihydrolase test:

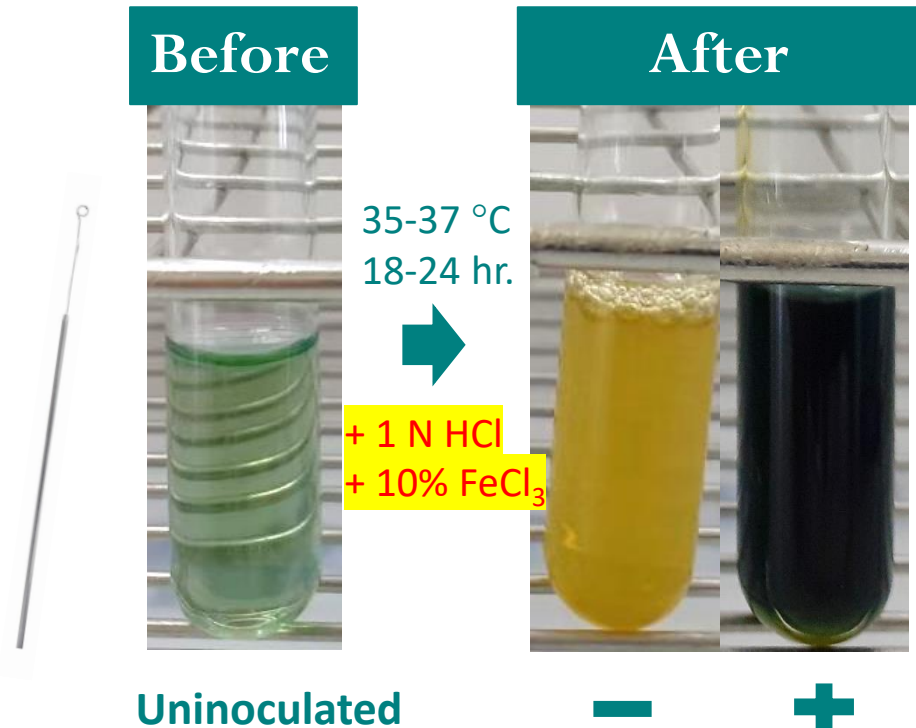
- Lysine decarboxylase (LD)
- Ornithine decarboxylase (OD)
- Arginine dihydrolase (AD)



3.1 Deaminase test

Phenylalanine deaminase (PD): For **gram negative bacilli**

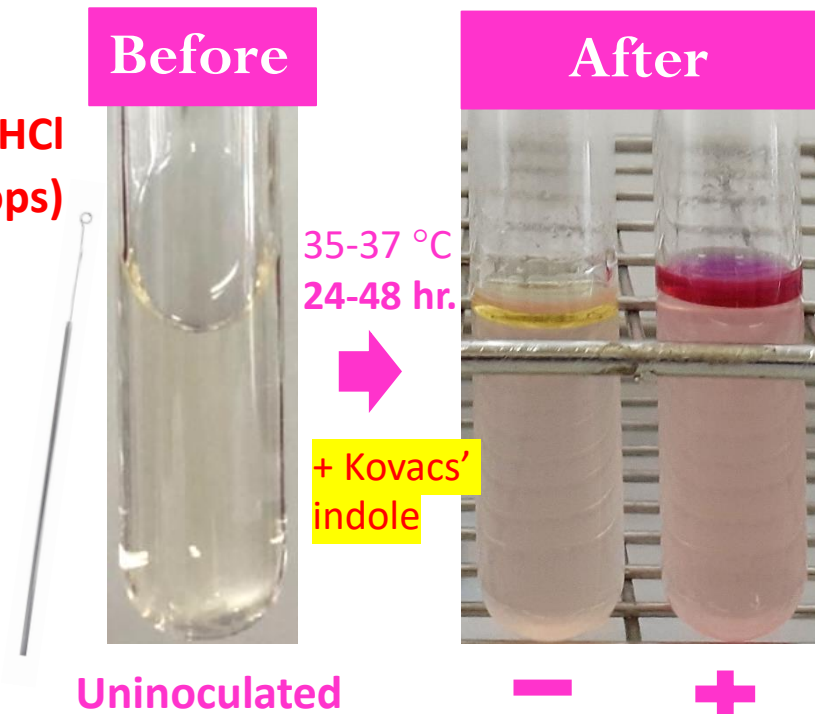
- ❖ **Purpose:** To determine the ability of bacteria to deaminate phenylalanine to phenylpyruvic acid which will react with ferric chloride and change to green color
- ❖ **Principle:** Phenylalanine → Phenylpyruvic acid + Ammonia
- ❖ Phenylpyruvic acid + FeCl_3 → green complex
- **Main components:** Malonate-PD broth + Phenylalanine *(combination test)
- **Reagent test:**
 - + 1 N HCl (1 drop) → turn to yellow
 - + 10% FeCl_3 (4-5 drops)
- **Interpretation:**
 - Positive:** Change from yellow to dark green (color fade in 3-5 min)
 - Negative:** Not change (yellow)



3.1 Deaminase test

Indole test : For **gram negative bacilli**, etc.

- ❖ **Purpose:** To determine the ability of bacteria to deaminate tryptophan to indole by tryptophanase
- ❖ **Principle:** Tryptophan → Indole + Pyruvate + Ammonia
Indole + p-dimethylaminobenzaldehyde (DMAB) → Red-violet
- **Main components:** Tryptone broth
- **Reagent test:** 5% DMAB in isoamyl alcohol-HCl (Kovacs' indole reagent) (2 drops)
store in brown bottle at 4°C
- **Interpretation:**
Positive: Red-violet ring on surface
Negative: Yellow ring on surface



3.2 Decarboxylase/Dihydrolase test

LD, OD, AD test : For fermentative bacteria (**bacilli**)

❖ **Purpose:** To determine ability of bacteria to decarboxylate specific amino acid to alkaline diamine products in acid and anaerobic condition

❖ **Principle:**

Glucose fermentation -> **Acid**

Lysine decarboxylase

Decarboxylation: 1. **LD:** L-Lysine → **Cadaverine**

Ornithine decarboxylase

2. **OD:** L-Ornithine → **Putrescine**

Arginine dehydrolase

3. **AD:** L-Arginine → L-Citrulline → L-Ornithine → **Putrescine**

- **Amino acid:** Lysine, Ornithine, Arginine
- **Main components:** basal medium, Glucose, **Amino acid**, Pyridoxyl, * + 1% NaCl (for *Vibrio* spp.)
- **Indicator:** **Bromocresol purple**
- **Procedure:** Always included **Control tube** (no amino acid) and add paraffin oil after inoculate

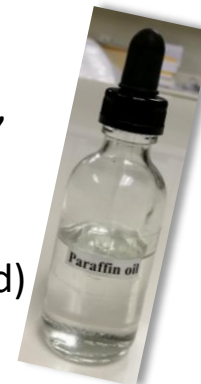
Control

Test



- No amino acid
- Glucose

- Amino acid
- Glucose



3.2 Decarboxylase/Dihydrolase test

LD, OD, AD test : For fermentative bacteria (**bacilli**)

➤ Interpretation: Control: must be **Yellow** (Indicator = Bromocresol purple)

Test: Positive: **Purple**

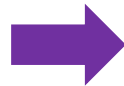
Negative: **Yellow** as same as control

Before

No amino acid
Glucose

Amino acid
Glucose

35-37 °C
24-48 hr.



PASS

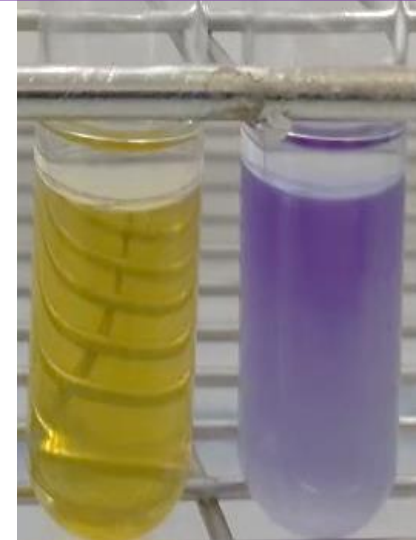
After



Control Test



Control



— +
|-----Test-----|



4. Enzyme production

- 4.1 Catalase test
- 4.2 Coagulase test
- 4.3 Bile esculin hydrolysis test
- 4.4 PYR test
- 4.5 Oxidase test
- 4.6 Urease test
- 4.7 Nitrate reduction test

4.1 Catalase test

Slide catalase test: For Gram positive cocci/bacilli, etc.

ใช้หลอด catalase ไม้

❖ **Purpose:** To detect catalase enzyme production

❖ **Principle:**
$$2\text{H}_2\text{O}_2 \xrightarrow{\text{catalase}} 2\text{H}_2\text{O} + \text{O}_2$$

➤ **Reagent test:** 3% Hydrogen peroxide (H_2O_2)
(1 drop), store in brown bottle at 4°C

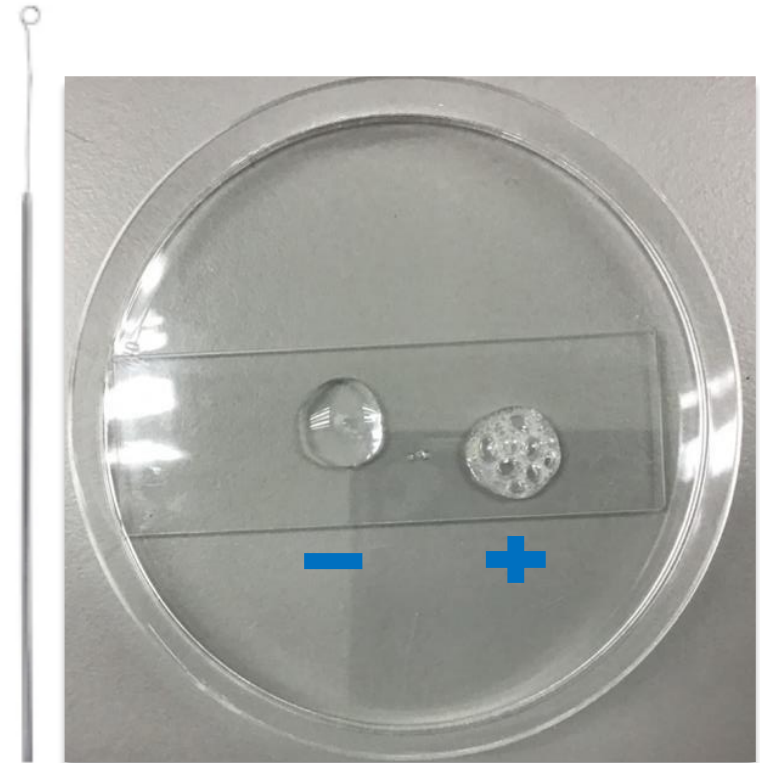
➤ **Interpretation:**

Positive: Many bubbles produced

Negative: No or very few bubbles produced

➤ **Precautions:** !

- Colony from BA -> FP
- Iron-containing inoculating loops -> FP



4.2 Coagulase test

Coagulase tube test: For *Staphylococcus* spp.

เช็ดกัก can ปล่อย fibrinogen เป็น fibrin ผ่าน การสร้าง Coagulase ได้ไหม

- ❖ **Purpose:** To determine the ability of bacteria to convert fibrinogen to fibrin by coagulase enzyme production

coagulase

- ❖ **Principle:** Fibrinogen → fibrin clot

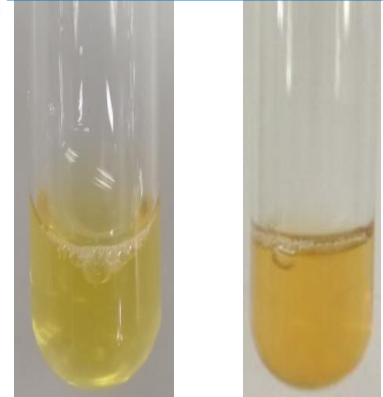
- **Main components:** EDTA-rabbit plasma, sodium fluoride- or EDTA-human plasma + TSB broth = 1:1
- **Procedure:** add 1 full loop of bacteria -> incubate at 35-37 °C
- **Interpretation:**

at 4hr. and at 18-24 hr. -> Grading clot (+1 to +4)

Positive: Fibrin clot (3+ to +4)

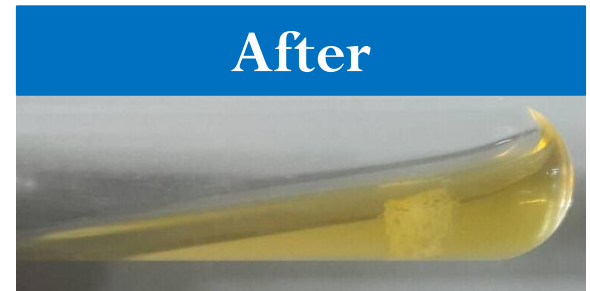
Negative: not change or Fibrin clot (1+ to +2)

Before



Uninoculated ↓ Inoculated

After



+



4.3 Bile esculin hydrolysis test

Bile esculin hydrolysis test: For *Streptococcus* & *Enterococcus* spp.

စို့ဘဲ Bile can ရှိသော Esculinase ရှိရှိသော 10-40% Bile ရှိရှိသော

- ❖ **Purpose:** To determine the ability of bacteria to produce esculinase enzyme and tolerance to 10-40% bile

- ❖ **Principle:** Esculinase
Esculin → esculetin + glucose
Esculetin + ferric citrate → **phenolic iron complex**

- **Main components:** basal medium, Esculin, Bile Salts, Ferric citrate, agar → solid agar

- **Interpretation:**

Positive: color change to **Black**

Negative: not change

Before



Uninoculated

35-37 °C
18-24 hr.



After



—

+

4.4 PYR test

Pyrrolidonyl Arylamidase (PYR) test: For group A Streptococcus & Enterococcus spp.

- ❖ **Purpose:** To determine the ability of bacteria to produce Pyrrolidonyl arylamidase (Pyrrolidonyl aminopeptidase)

L-pyrrolidonyl arylamidase

- ❖ **Principle:** L-pyrrolidonyl- β -naphthylamide $\rightarrow$ β -naphthylamide
 β -naphthylamide + DMACA $\rightarrow$ Red color (Schiff base)

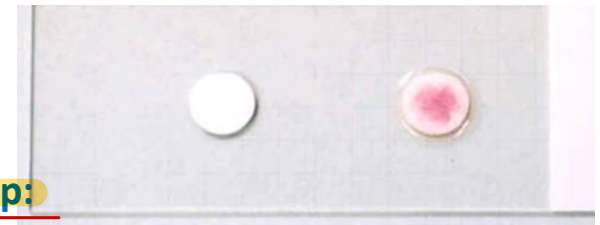
- **Main components:** basal medium, 0.01-0.02% L-pyrrolidonyl- β -naphthylamide $\rightarrow$ PYR agar / **PYR broth** / **PYR strip**
- **PYR reagent:** 0.01% p-dimethylaminocinnamaldehyde (DMACA), store in brown bottle at 4°C
- **Interpretation:**

Allow up to 1 to 2 minutes for a color to develop.

Positive: color change to Red

Negative: not change (or other colors)

PYR strip:



PYR broth:



4.5 Oxidase test

Oxidase test: For Gram negative bacilli/cocci

❖ **Purpose:** To detect the presence of cytochrome c oxidase enzyme

cytochrome c oxidase

❖ **Principle:** $\text{TMPD} + \text{O}_2 \rightarrow \text{Indophenols} + \text{H}_2\text{O}$
(colorless) (purple or dark blue color)

➤ **Reagent test:**

1% Tetramethyl-p-phenylenediamine dihydrochloride
(TMPD, Kovács oxidase reagent)

, store in brown bottle at 4°C

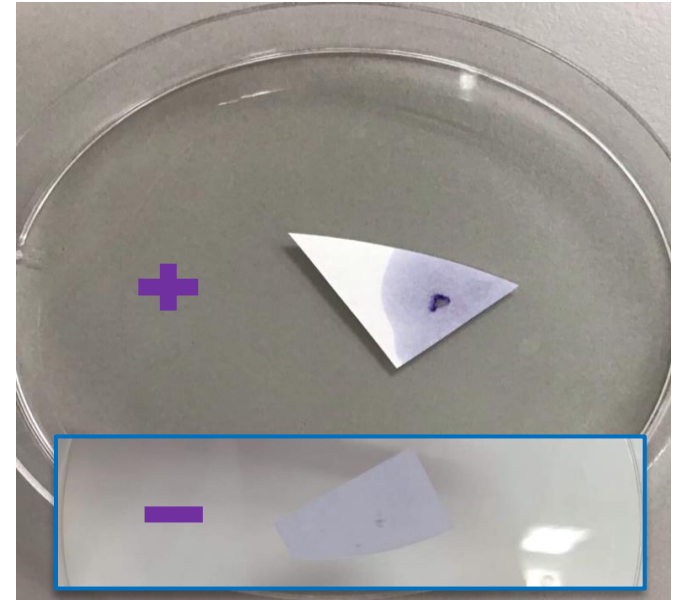
➤ **Interpretation:**

Positive: color changes to dark purple within 5 to 10s

Negative: not change or it takes longer

➤ **Precautions:** Use fresh reagents and fresh culture

- Older cultures -> FN
- Acid-producing colony -> FN
- Iron-containing inoculating loops -> FP



4.6 Urease test

Urease test: For Gram negative bacilli, etc.

ตรวจหาการผลิต Urease ไข่

❖ **Purpose:** To detect urease production

Urease

❖ **Principle:** $\text{urea} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{ammonia}$

➤ **Main components:**

Christensen's urea agar
(2% **urea**, peptone, agar → solid agar)

➤ **Indicator:** **Phenol red**

➤ **Interpretation:**

Positive: color change to **Pink-red**

Negative: not change

Before



Uninoculated

35-37 °C
4-24 hr.



After



—

+

4.7 Nitrate reduction test

Nitrate reduction test: For **gram negative bacilli**, etc.

❖ **Purpose:** To determine the ability of bacteria to reduce nitrate (NO_3) to reduced product by nitrate reductase

❖ **Principle:**

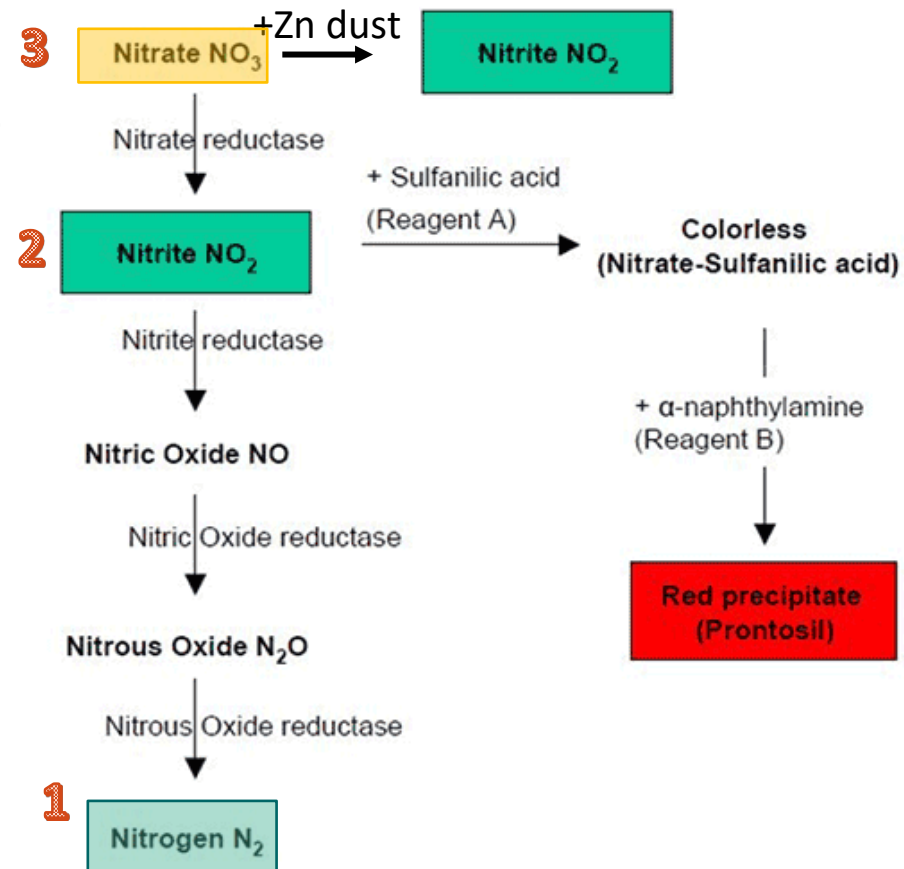
➤ **Main components:** Basal m., Potassium nitrate

➤ **Reagent test:**

- **Solution A: 0.8% sulfanilic acid (5 drops)**
- **Solution B: 0.5% α -naphthylamine (5 drops)**

Zn dust

(catalyst for remain $\text{NO}_3 \rightarrow \text{NO}_2$)



4.7 Nitrate reduction test

➤ Interpretation:

Positive:

1. **Gas** in Durham tube
2. Add solution A+B → **Red**
3. If add A+B → no color
→ add Zn dust → **No color**

Negative:

1. No gas in Durham tube
2. If add A+B → no color
→ add Zn dust → **Red**

Before



Uninoculated

35-37 °C
18-24 hr.

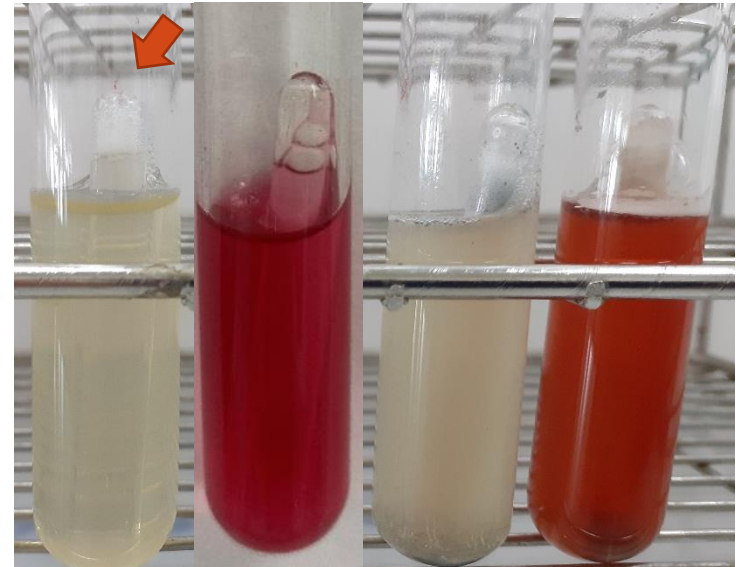


After

Sol. A+B

Sol. A+B
Zn dust

Sol. A+B
Zn dust



+

+

+

-

$\text{NO}_3^- \rightarrow \text{N}_2$

$\text{NO}_3^- \rightarrow \text{NO}_2$

NO_3^-
→ Other

NO_3^-

5. Tolerance test

- 5.1 6.5% Sodium chloride (NaCl) test
- 5.2 Growth in NaCl-peptone water
- 5.3 Growth at 42°C

“Avoid using a large inoculum that may cause false positive result”

5.1 6.5% Sodium chloride (NaCl) test

6.5% NaCl test : For *Streptococcus & Enterococcus spp.*

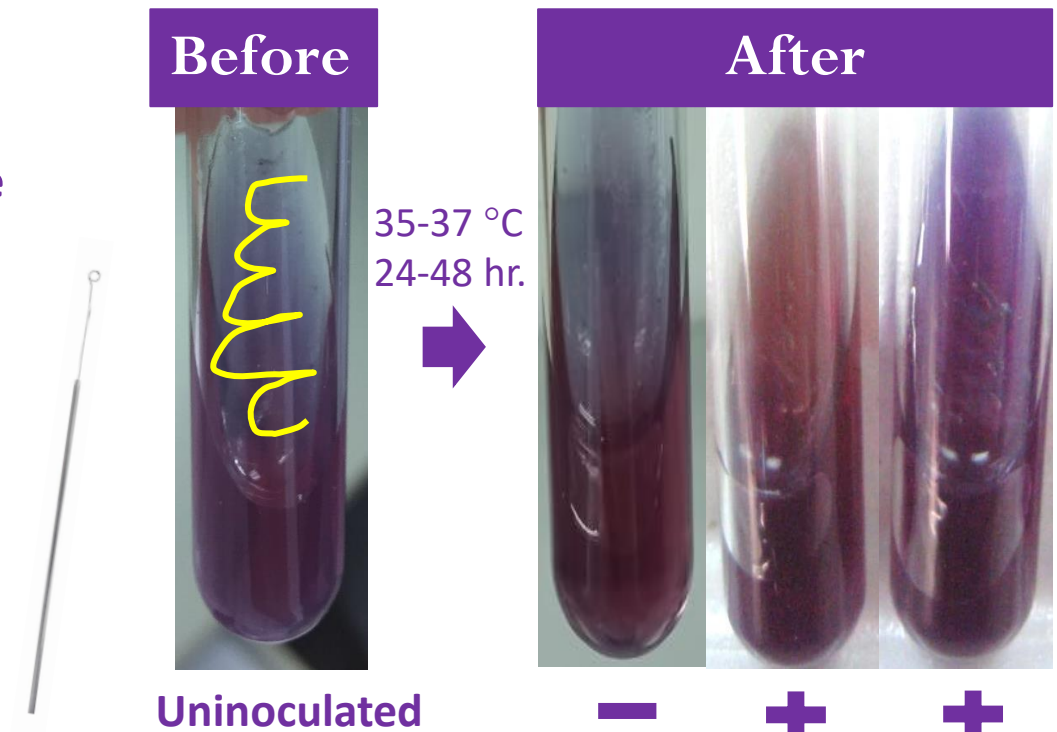
- ❖ **Purpose:** To determine the salt tolerance ability of bacteria by growth on 6.5% NaCl medium
- ❖ **Principle:** Growth on 6.5% NaCl medium → Use glucose → acid

- **Main components:**
NaCl, basal medium, glucose
- **Indicator:** Bromocresol purple
- **Interpretation:**

Positive: Growth

Negative: No growth

*Large inoculum -> FP



5.2 Growth in NaCl-peptone water

- ❖ **Purpose:**
- To differentiate ***Vibrio* spp. and *Aeromonas* spp.**
 - To determine the salt tolerance ability of bacteria by growth on different concentration of NaCl

❖ **Principle:** growth on different percentage of NaCl-peptone water → **Turbidity**

➤ **Main components:**

Peptone water with NaCl
0%, 1%, 3%, 6%, 8%, 10%

➤ **Interpretation:**

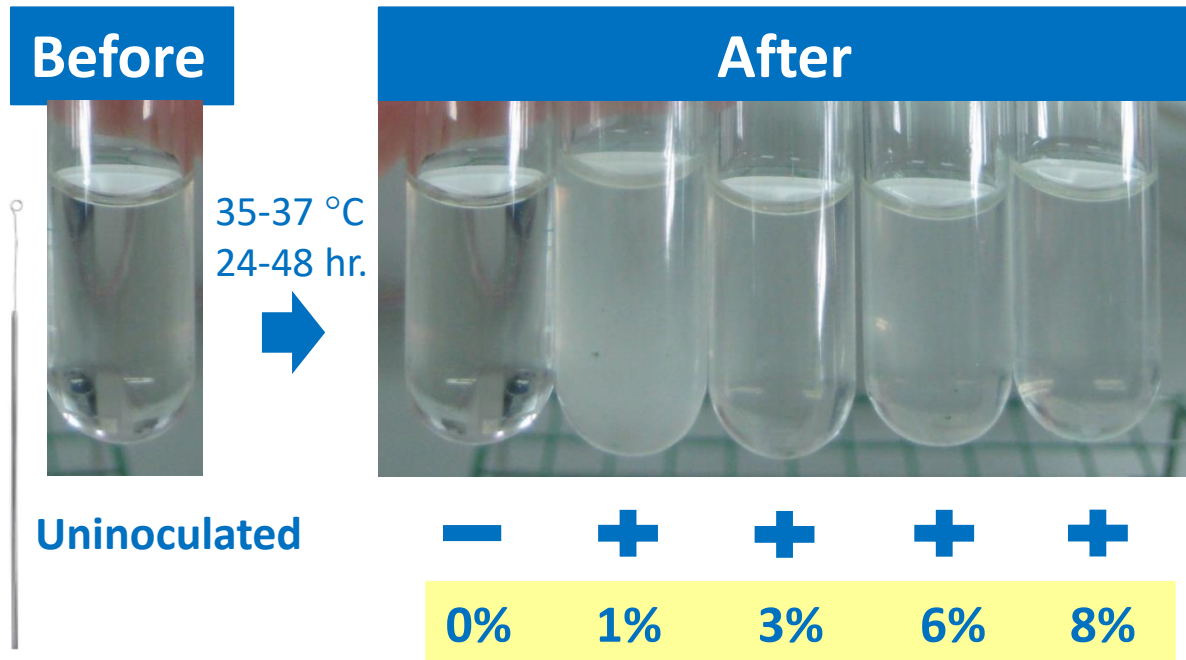
Positive: Turbid

Negative: Not Turbid

➤ **Precautions:**

- Direct colony inoculation
should not be used

*Large inoculum -> FP



5.3 Growth at 42°C

Growth at 42°C: For gram negative bacilli -> NFB

❖ **Purpose:** To determine the thermo tolerance ability of bacteria

❖ **Principle:** **Growth at 42°C**

➤ **Main components:**

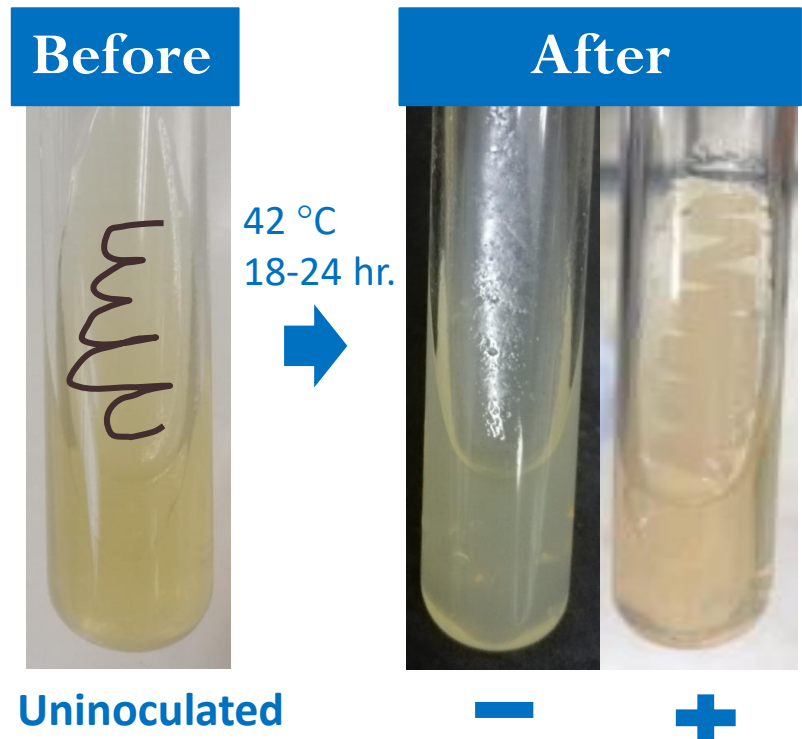
Tryptic Soy Agar (TSA) or nutrient agar

➤ **Interpretation:**

Positive: Growth

Negative: No growth

*Large inoculum -> FP



6. Susceptibility test

- **Purpose:** To determine antimicrobial susceptibility against bacteria (gram positive cocci)

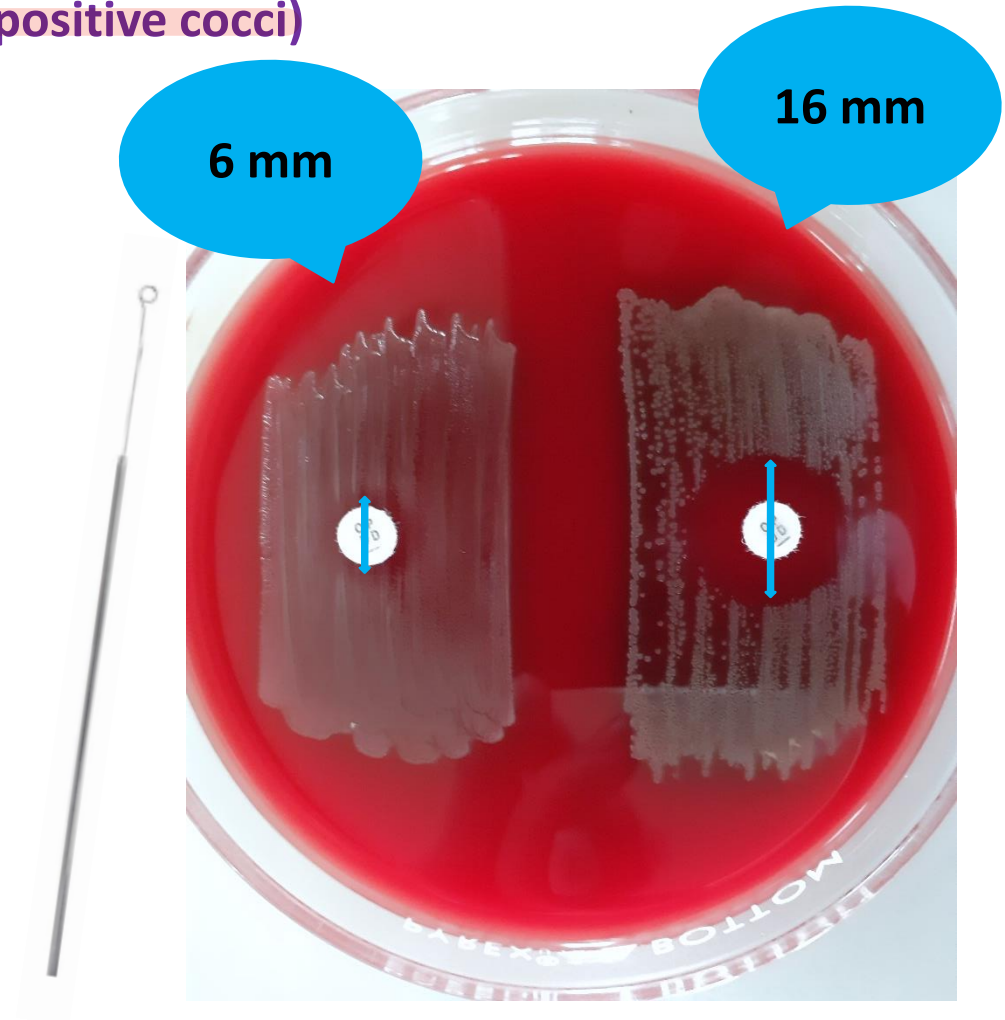
6.1 Novobiocin

6.2 Bacitracin

6.3 Optochin

➤ **Interpretation:**

- Resistant (R)
- Susceptible (S) or Sensitive (S)



6.1 Novobiocin susceptibility test

❖ Purpose:

- To differentiate Coagulase-negative staphylococci (CONS) and presumptively identify the isolate as *Staphylococcus saprophyticus*.

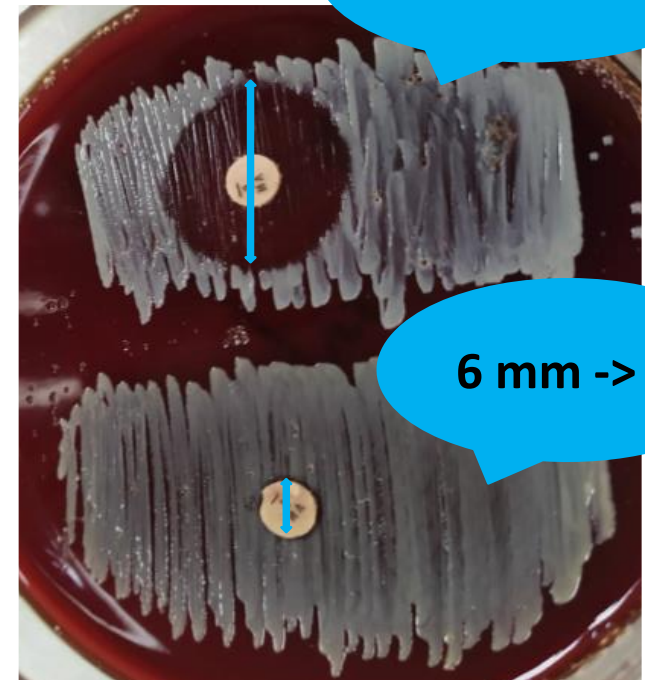
- ❖ Principle: Novobiocin (NV, 5 µg) can inhibit most of CONS,
except *S. saprophyticus* (novobiocin resistant)

- Media: Sheep blood agar
- Condition: incubate at 35-37 °C for 18-24 hr.

➤ Interpretation:

Susceptible (S): inhibition zone > 16 mm

Resistant (R): inhibition zone ≤ 16 mm



6.2 Bacitracin susceptibility test

❖ Purpose:

- To differentiate Beta-hemolytic *Streptococcus* and presumptively identify the isolate as **Group A beta-hemolytic *Streptococcus***
- To differentiate *Micrococcus* spp. from *Staphylococcus* spp.

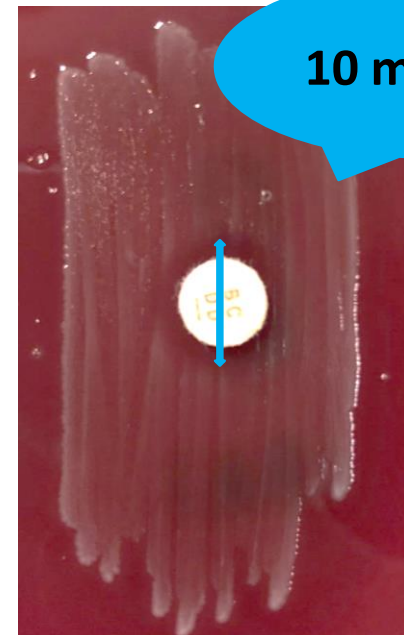
❖ Principle: Bacitracin (BC, 0.04 units) can inhibit Group A beta-hemolytic *Streptococcus* and *Micrococcus* spp.

- Media: Sheep blood agar
- Condition: incubate at 35-37 °C for 18-24 hr.

➤ Interpretation:

Susceptible (S): inhibition zone > 6 mm

Resistant (R): inhibition zone ≤ 6 mm



6.3 Optochin susceptibility test

❖ **Purpose:** To differentiate Alpha-hemolytic *Streptococcus* and presumptively identify the isolate as *Streptococcus pneumoniae*

❖ **Principle:** Optochin (OP, 5 µg) can inhibit *S. pneumoniae*

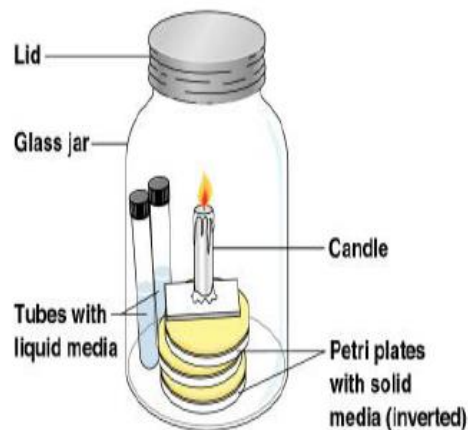
➤ **Media:** Sheep blood agar

➤ **Condition:** incubate at 35-37 °C for 18-24 hr.
in candle jar (5% CO₂)

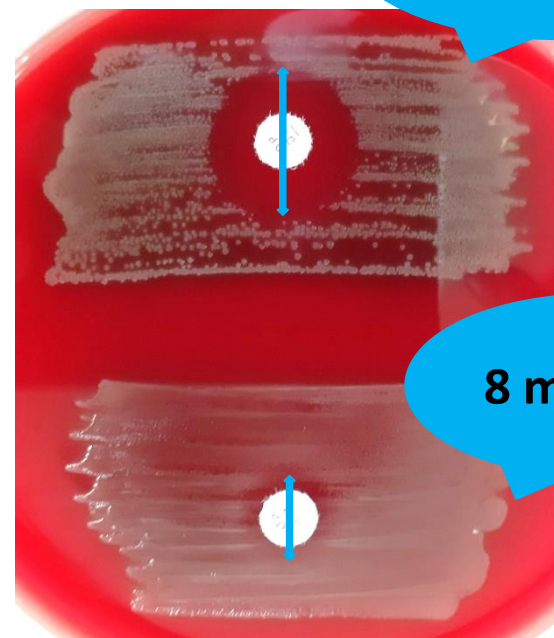
➤ **Interpretation:**

Susceptible (S): inhibition zone ≥ 14 mm

Resistant (R): inhibition zone < 14 mm



Candle jar



18 mm -> S

8 mm -> R

7. Miscellaneous tests

7.1 Pigment production:

- PPR
- King P / King F

7.2 Motility test

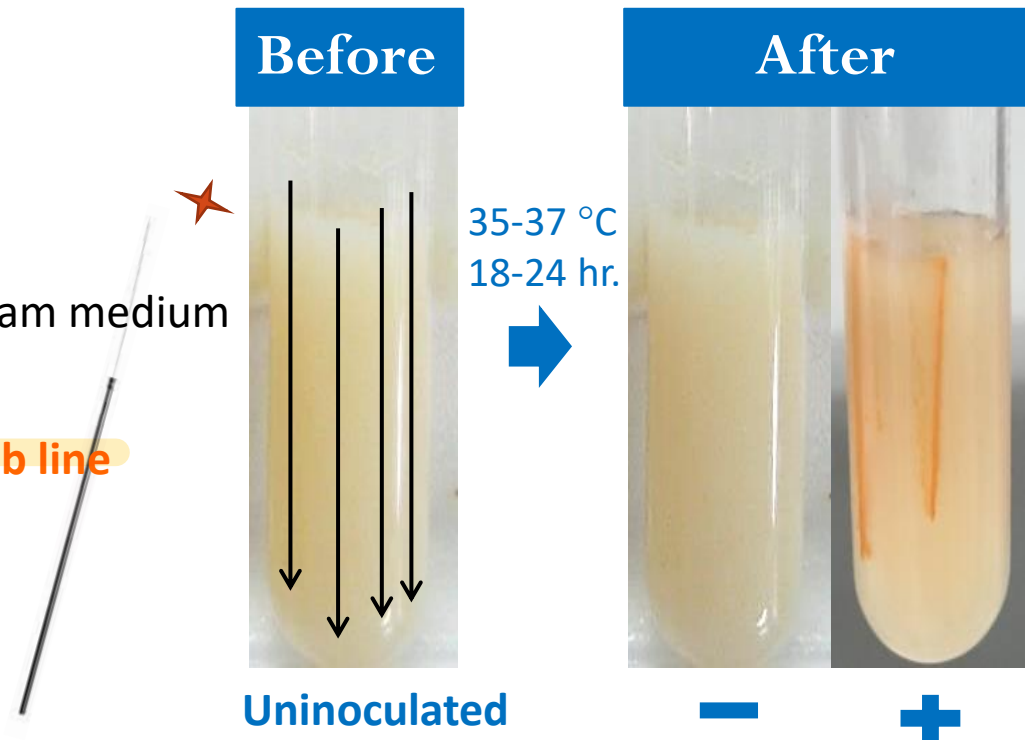
7.3 Combination test:

- MIO (Motile/Indole/OD)
- Lysine Iron Agar (LIA) Test

7.1 Pigment production

Proteose Peptone Rice agar test (PPR): *Streptococcus*

- ❖ **Purpose:**
 - To determine the ability of pigment-producing under specific condition
 - To differentiate Group B beta-hemolytic *Streptococcus* from other
- ❖ **Principle:** Gr. B *Streptococcus* can produce the orange pigment on starch media under anaerobic condition
- **Main components:**
 - Peptone and Proteose Peptone Rice (Milintangura S.)
 - Replace for Granada medium and Islam medium
- **Interpretation:**
 - Positive:** Orange pigment along stab line
 - Negative:** No change



7.1 Pigment production

King P/F test: For **NFB** -> ***Pseudomonas* spp.**

❖ Purpose:

- To determine the ability of a particular pigment producing
- To differentiate some *Pseudomonas* spp.

❖ Principle:

- **King P** → **Pyocyanin (blue-green)**
- **King F** → **Fluorescein/pyoverdinin (Yellow-green)**

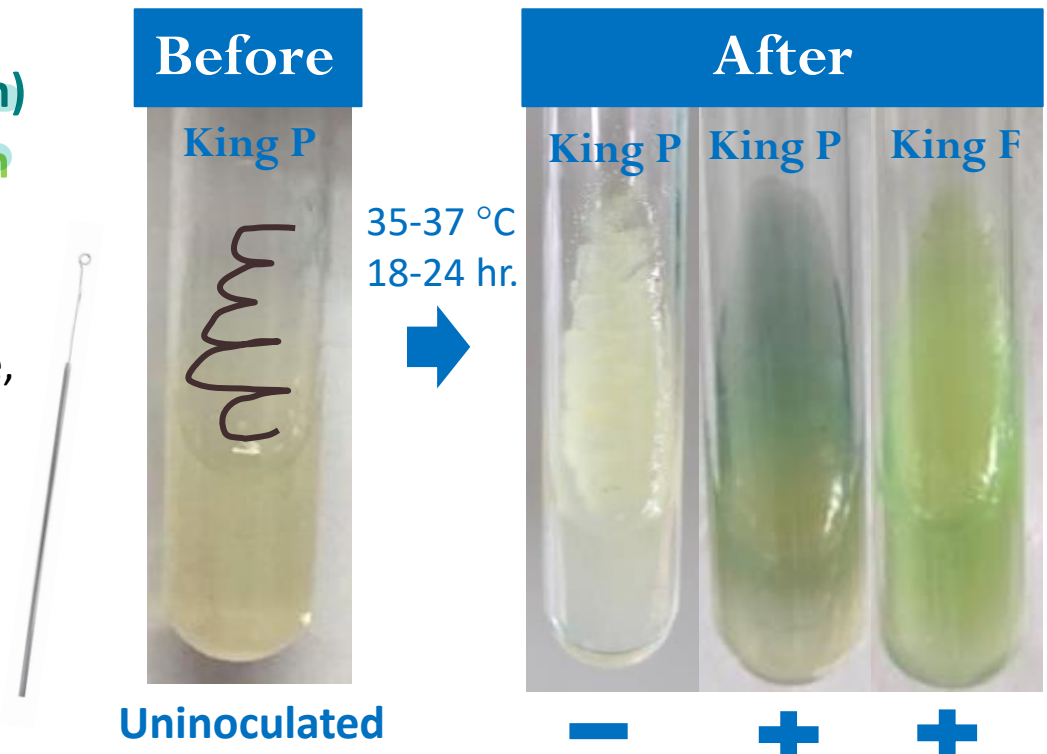
➤ Main components:

Basal medium, magnesium, phosphate, glycerol and **no iron!**

➤ Interpretation:

Positive: **Growth with color pigment**

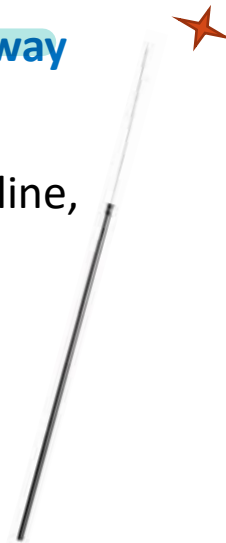
Negative: Growth with no pigment



7.2 Motility test

Motility test : For **gram negative bacilli**, etc.

- ❖ **Purpose:** To determine the motility of bacteria
- ❖ **Principle:** some bacteria can be motile in semisolid medium by flagella
- **Main components:** 0.4% agar -> Semi-solid agar
- **Procedure:** 1 stabbing in the center by needle
- **Interpretation:**
 - Positive:** Turbid area extending away from stab line
 - Negative:** Growth with sharp stab line, not turbid



Before

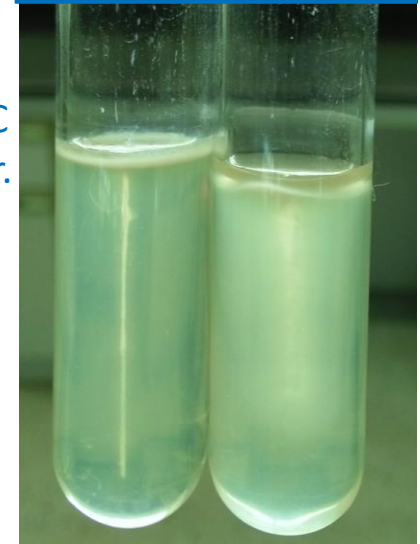


Uninoculated

35-37 °C
18-24 hr.



After



—

+

7.3 Combination test

MIO (Motile, Indole, OD): For fermentative negative bacilli

- **Principle:** Combine test of motility (M), indole (I) and ornithine decarboxylase tests (O)
- **Main components:** 0.4% agar -> Semi-solid agar, Casein peptones, 0.5% ornithine, 1% glucose
- **Indicator:** Bromocresol purple
- **Procedure:** 1 stabbing in the center by needle
- **Interpretation:**

1. Motile:

positive → Turbid area extending away from stab line

negative → Growth with sharp stab line, not turbid

2. OD:

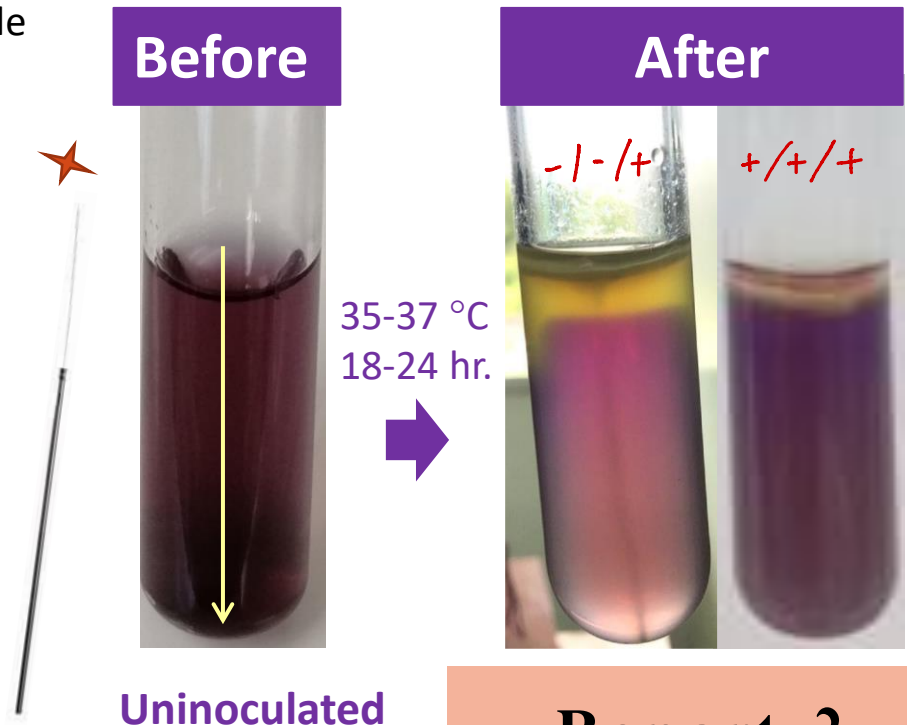
positive → Purple at butt

negative → yellow at butt

3. Indole: Kovacs' indole reagent (3-4 drops)

positive → Red ring

negative → yellow ring



7.3 Combination test

Lysine Iron Agar (LIA) Test: For fermentative gram negative bacilli

- **Principle:** Combine test of Lysine deaminase, Lysine decarboxylase (LD) and H_2S
- **Indicator:** Bromocresol purple
- **Procedure:** streak and stab by needle
- **Interpretation:** R-K-A

1. Lysine deaminase ≈ PD:

positive → Red at slant

negative → No change

2. LD:

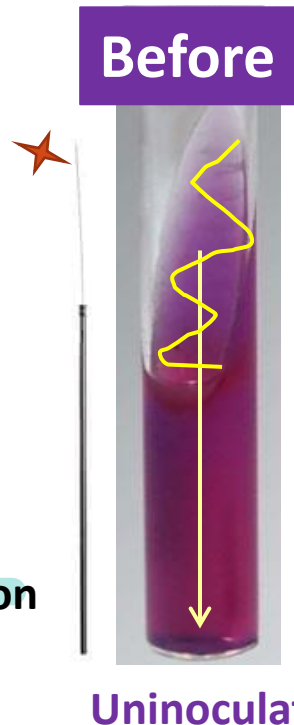
positive → Purple at butt

negative → Yellow at butt

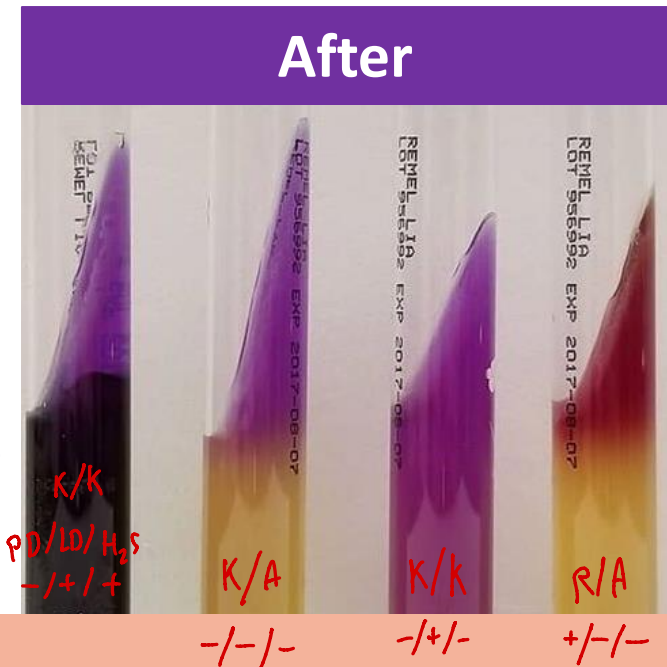
3. H_2S :

positive → black precipitation

negative → No change



35-37 °C
18-24 hr.



Report ?

การควบคุมคุณภาพของอาหารเลี้ยงเชื้อ: หลักการ



ควรทำหลักการควบคุมคุณภาพทุกครั้งที่มีการเปิดใช้ Media lot ใหม่ หรือเตรียมใหม่

1. ทดสอบสภาพของอาหารก่อนใช้งาน (Gross physical appearance)

- ดูสภาพและลักษณะของอาหารด้วยตาเปล่า

2. ทดสอบการปลอดเชื้อ (Sterility test)

- สุ่ม 5-10% ของอาหารเลี้ยงเชื้อ -> incubate at 35-37°C for 18-24 hr. + at RT 48 hr.

3. ทดสอบคุณสมบัติของอาหาร (Performance test/ Usage control)

- ทดสอบความสามารถในการเจริญเติบโตและการแสดงคุณสมบัติการใช้งานของอาหาร
- เชื้อมาตรฐาน ATCC
- วิธีให้เหมาะสมกับอาหารทดสอบ

A

ชื่ออาหารเพาะเชื้อ	เชื้อที่ใช้ทดสอบ	ปฏิกิริยา
Blood agar	<i>Streptococcus pneumoniae</i>	Alpha-hemolysis
	Group A streptococci	Beta-hemolysis
Chocolate agar	<i>Haemophilus influenzae</i>	เจริญเติบโตดี
	<i>Neisseria gonorrhoeae</i>	เจริญเติบโตดี
MacConkey agar	<i>Escherichia coli</i>	โคโลนีสีชมพู
	<i>Proteus mirabilis</i>	โคโลนีไม่มีสี
	Gram positive strain	ไม่สามารถเจริญเติบโต
Voges-Proskauer	<i>Klebsiella pneumoniae</i>	สีแดง (+)
	<i>Escherichia coli</i>	ไม่มีสี (-)
Lysine decarboxylase	<i>Klebsiella pneumoniae</i>	สีม่วง (+)
	<i>Enterobacter cloacae</i>	สีเหลือง (-)
Ornithine decarboxylase	<i>Enterobacter cloacae</i>	สีม่วง (+)
	<i>Klebsiella pneumoniae</i>	สีเหลือง (-)
Arginine dihydrolase	<i>Enterobacter cloacae</i>	สีม่วง (+)
	<i>Klebsiella pneumoniae</i>	สีเหลือง (-)
Nitrate broth	<i>Escherichia coli</i>	สีแดง (+) หลังจากหยด Sol ⁿ A และ B
	<i>Pseudomonas aeruginosa</i>	ไม่มีสี (+) หลังจากหยด Sol ⁿ A และ B และผงสังกะสี
	<i>Acinetobacter baumannii</i>	สีแดง (-) หลังจากหยด Sol ⁿ A และ B และผงสังกะสี
6.5% NaCl broth	<i>Enterococcus faecalis</i>	สีเหลือง (+)
	Group A streptococci	สีม่วง (-)

C

D

D

C

ชื่ออาหารเพาะเชื้อ	เชื้อที่ใช้ทดสอบ	ปฏิกิริยา
Bile-esculin agar	<i>Enterococcus faecalis</i>	สีดำ (+)
	Other streptococci, not group D	ไม่มีสี (-)
Urea agar	<i>Proteus mirabilis</i>	สีชมพู (+)
	<i>Escherichia coli</i>	สีเหลือง (-)
Simmon citrate agar	<i>Klebsiella pneumoniae</i>	สีน้ำเงิน (+)
	<i>Escherichia coli</i>	สีเขียว (-)
Triple sugar iron slant (TSI)	<i>Salmonella</i> group	K/A H ₂ S
	<i>Shigella</i> spp.	K/A
	<i>Escherichia coli</i>	A/AG
	<i>Pseudomonas aeruginosa</i>	K/K
Motility test medium (0.4% agar)	<i>Escherichia coli</i>	ขุ่น (+)
	<i>Klebsiella pneumoniae</i>	เชื้อขึ้นเฉพาะที่รอย stab และวุ้นใส (-)
DNase agar	<i>Staphylococcus aureus</i>	Clear zone (+)
Indole	<i>Escherichia coli</i>	สีแดง (+)
	<i>Klebsiella pneumoniae</i>	ไม่มีสี (-)
Mannitol broth	<i>Escherichia coli</i>	สีฟ้า (+)
	<i>Plesiomonas shigelloides</i>	สีเขียว (-)

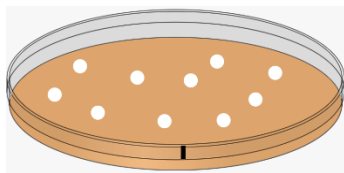
Method of inoculation for QC

Colony suspension

Control strain



Direct colony

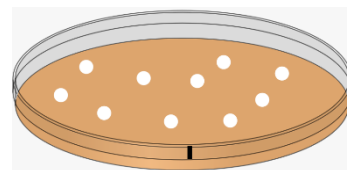


D



0.5 McFarland:

10 μ l



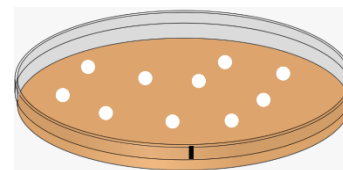
C

dilute 1:10



Dilution 1:10

10 μ l



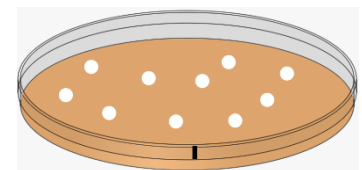
B

dilute 1:10



Dilution 1:100

10 μ l



A

การควบคุมคุณภาพของอาหารเลี้ยงเชื้อ: เอกสาร QC



ตัวอย่างตารางบันทึกการควบคุมคุณภาพอาหารเลี้ยงเชื้อ และน้ำยา

Medium (plate): _____

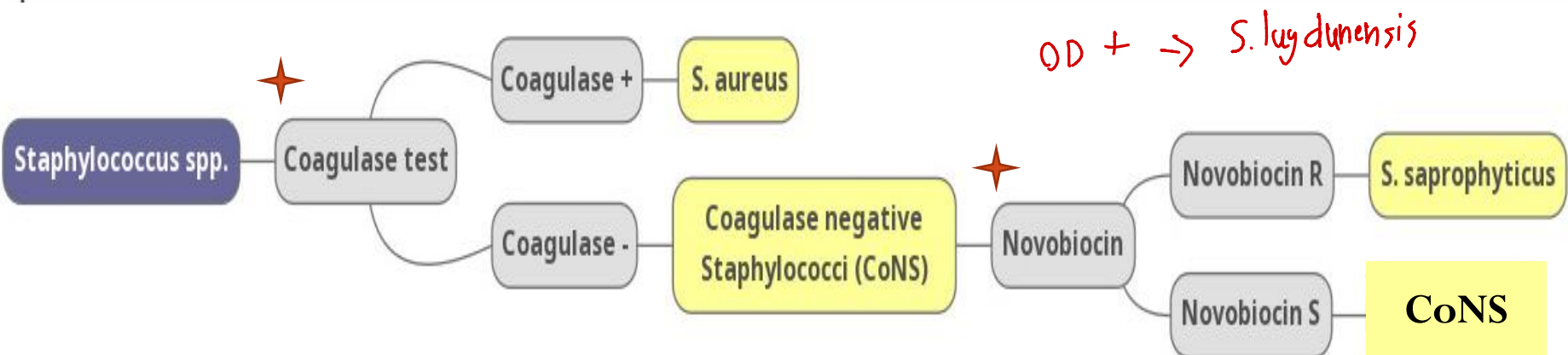
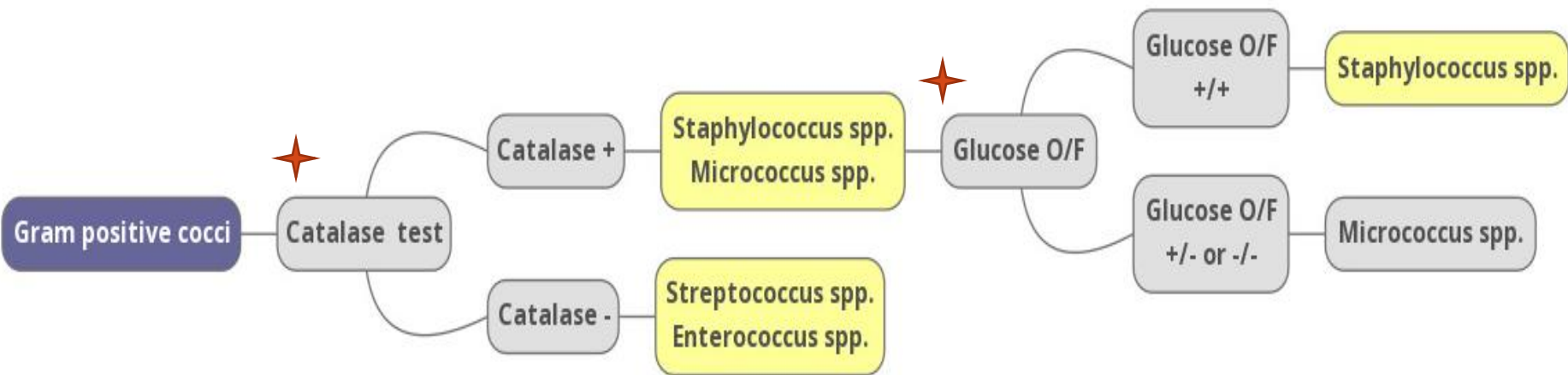
Medium (tube): _____

Reagent: _____

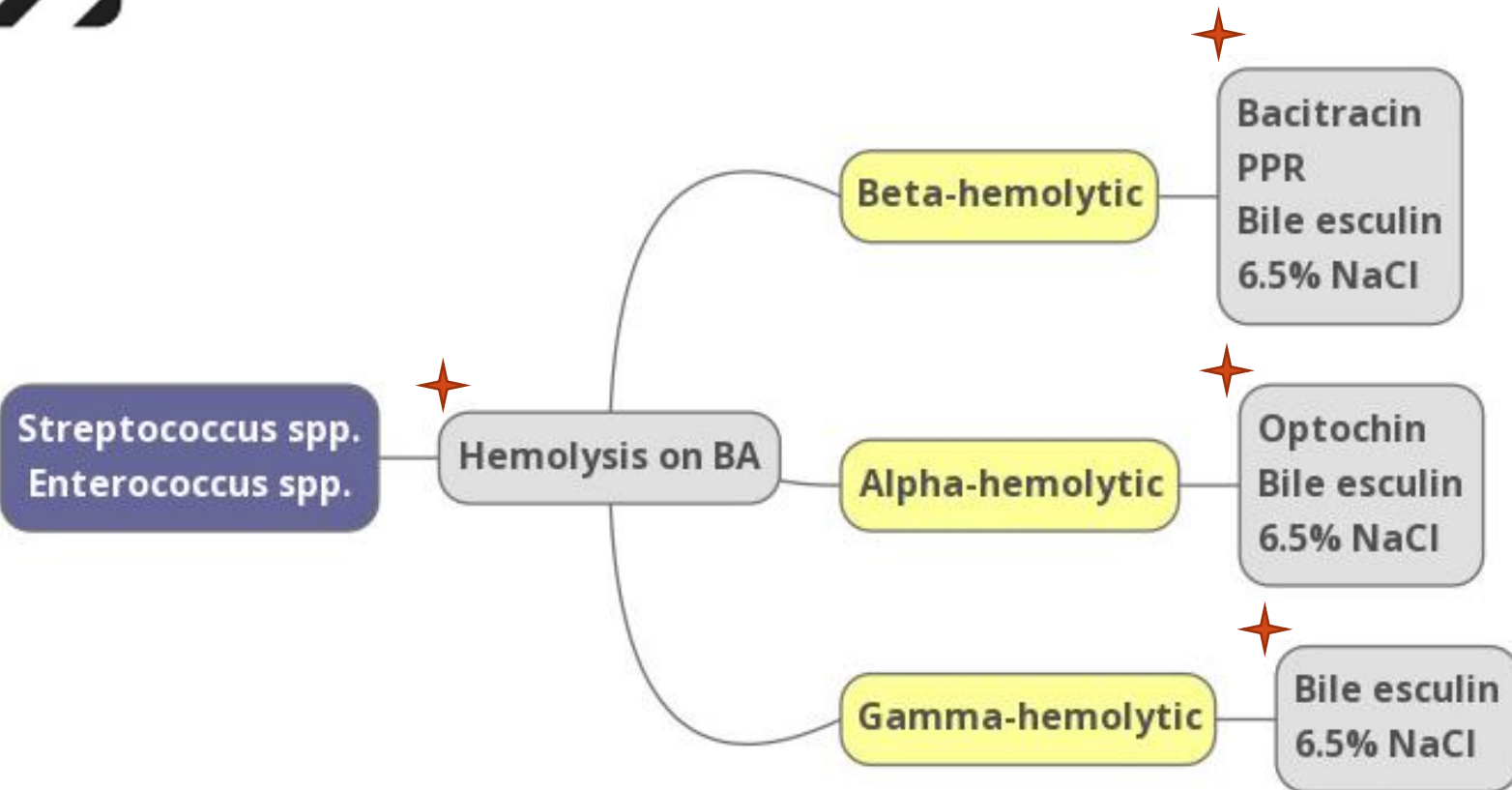
Interpretation*	Date prepared	Source	Lot No.	Positive control	Reaction	Negative control	Reaction	Date performed	Checked by

* A = Acceptable และ NA = Non-acceptable

Identification of Gram positive cocci



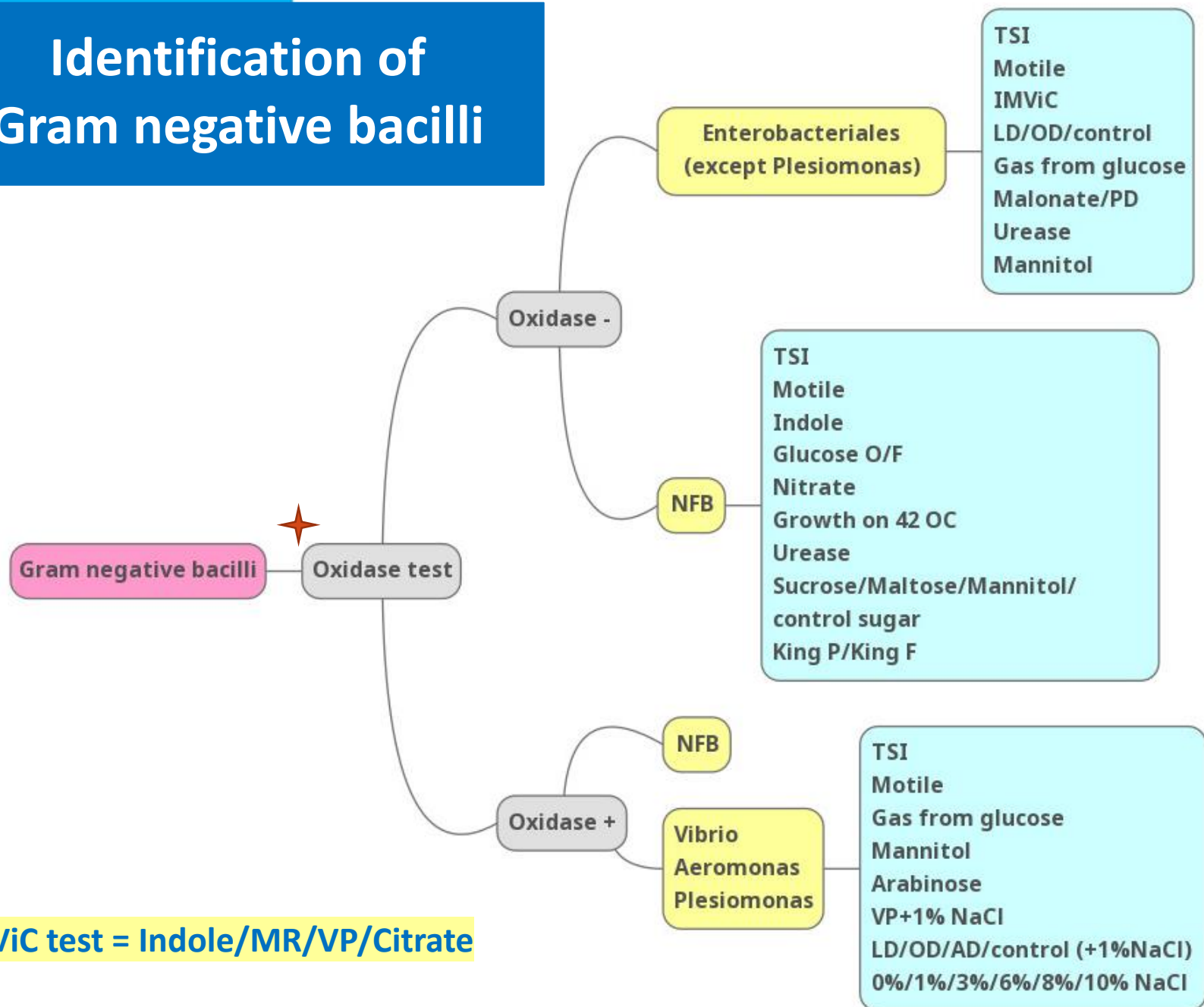
Identification of Gram positive cocci



Identification of *Streptococcus* spp.

Bacteria	Bacitracin	Optochin	PPR	Bile esculin	6.5% NaCl
Beta-hemolysis					
Group A <i>Streptococcus</i>	S		-	-	-
Group B <i>Streptococcus</i>	R		+	-	-
β-<i>Streptococcus</i> Group D	R		-	+	-
<i>Streptococcus</i> not gr. A, B, D	R		-	-	-
β-<i>Enterococcus</i> spp.	R		-	+	+
Alpha-hemolysis					
<i>Streptococcus pneumoniae</i>		S		-	-
Viridans <i>Streptococcus</i>		R		-	-
α-<i>Streptococcus</i> Group D		R		+	-
α-<i>Enterococcus</i> spp.		R		+	+
Gamma-hemolysis					
Viridans <i>Streptococcus</i>				-	-
γ-<i>Streptococcus</i> Group D				+	-
γ-<i>Enterococcus</i> spp.				+	+

Identification of Gram negative bacilli



*IMViC test = Indole/MR/VP/Citrate

เอกสารอ้างอิง

- หนังสือการทดสอบชีวเคมีที่ใช้ในการตรวจวินิจฉัยชนิดแบคทีเรียทางการแพทย์ พ.ศ. 2556 โดย ผศ.ดร. เกศรา วัดอักษร
- คู่มือมาตรฐานห้องปฏิบัติการจุลชีววิทยาทางการแพทย์และสาธารณสุข พ.ศ. 2560 สถาบันวิจัยวิทยาศาสตร์สาธารณสุข กรมวิทยาศาสตร์การแพทย์ กระทรวงสาธารณสุข
- คู่มือการปฏิบัติงานแบคทีเรียและรา สำหรับโรงพยาบาลศูนย์และโรงพยาบาลทั่วไป พ.ศ. 2558 โดยสถาบันวิจัยวิทยาศาสตร์สาธารณสุข กรมวิทยาศาสตร์การแพทย์ กระทรวงสาธารณสุข และ ศูนย์ความร่วมมือไทย-สหรัฐ ด้านสาธารณสุข
- Atlas R, Snyder J. Reagents, Stains, and Media: Bacteriology. Manual of Clinical Microbiology, 11th Edition, 2015. p. 316-49.
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- Becker K, Skov RL, von Eiff C. Staphylococcus, Micrococcus, and Other Catalase-Positive Cocci. Manual of Clinical Microbiology, 11th Edition, 2015. p. 354-82.
- Forsythe SJ, Abbott SL, Pitout J. Klebsiella, Enterobacter, Citrobacter, Cronobacter, Serratia, Plesiomonas, and Other Enterobacteriaceae. Manual of Clinical Microbiology, 11th Edition, 2015. p. 714-37.
- Tarr CL, Bopp CA, Farmer III JJ. Vibrio and Related Organisms. Manual of Clinical Microbiology, 11th Edition, 2015. p. 762-72.

สื่อการเรียนรู้เพิ่มเติม

- หนังสือการทดสอบชีวเคมีที่ใช้ในการตรวจวินิจฉัยชนิดแบคทีเรียทางการแพทย์ พ.ศ. 2556 โดย ผศ.ดร. เกศรา วัดอักษร
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(http://narst.dmsc.moph.go.th/manuals/standard_manual_2560.pdf)
- คู่มือการปฏิบัติงานแบคทีเรียและรา สำหรับโรงพยาบาลศูนย์และโรงพยาบาลทั่วไป พ.ศ. 2558 โดยสถาบันวิจัยวิทยาศาสตร์สาธารณสุข กรมวิทยาศาสตร์การแพทย์ กระทรวงสาธารณสุข และ ศูนย์ความร่วมมือไทย-สหรัฐ ด้านสาธารณสุข
(<http://narst.dmsc.moph.go.th/data/Idenbook.pdf>)
- Manual of Clinical Microbiology, 11th Edition
(<https://onlinelibrary.wiley.com/doi/book/10.1128/9781555817381>)
- American Society for Microbiology, Protocols
(<https://asm.org/Browse-By-Content-Type/Protocols>)
- Video สาคิต ประกอบการเรียน ในบทเรียนออนไลน์บนระบบ Google Classroom: MTMI 202 Basic Microbiology
Google Classroom Link: <https://classroom.google.com/c/NTIzMTE1NzI33MDEz?cjc=eftv22u5>
Google Classroom Code: eftv2u5

Thank You

